

38_Comoros

October 18, 2025

1 How have infant, under-five, and neonatal deaths evolved relative to one another in Comoros between 1980 and 2020?

1.1 Abstract

Using World Bank World Development Indicators (WDI), this study examines the evolution of infant, under-five, and neonatal mortality in Comoros between 1980 and 2020. The analysis compares three key indicators: number of infant deaths, number of under-five deaths, and number of neonatal deaths. Over this forty-year period, all three mortality indicators declined substantially and consistently, reflecting major improvements in child health outcomes. The reduction in deaths was particularly steep during the 1990s and early 2000s, coinciding with broader global efforts to improve maternal and child health. Although neonatal mortality declined more slowly relative to infant and under-five mortality, all categories showed significant progress by 2020. These patterns reflect sustained public health interventions, improved access to healthcare services, and progress toward global development goals focused on child survival.

1.2 1. Question

How have infant, under-five, and neonatal deaths evolved relative to one another in Comoros between 1980 and 2020?

- **Infant deaths proxy:** Number of infant deaths
- **Under-five deaths proxy:** Number of under-five deaths
- **Neonatal deaths proxy:** Number of neonatal deaths

1.3 2. Data

- **Source:** World Bank World Development Indicators (WDI)
- **Indicators:**
 - Number of infant deaths
 - Number of under-five deaths
 - Number of neonatal deaths
- **Coverage:** Comoros, 1980–2020
- **Notes:** National-level data only

1.4 3. Method

1. Filtered dataset for Comoros and selected the three child mortality indicators.
2. **Extracted relevant columns:** Year, Indicator Name, and Value.

3. Pivoted the dataset to create a chronological comparison of infant, under-five, and neonatal deaths between 1980 and 2020.
4. Produced a multi-line graph showing the relative decline across all three mortality categories over time.

(Analysis is descriptive; no causal inference applied.)

1.5 4. Results

- **Infant deaths:** Declined dramatically from the early 1980s to 2020, showing a sustained reduction in child mortality.
- **Under-five deaths:** Showed the steepest overall decline, reflecting broad improvements in early childhood survival and disease prevention.
- **Neonatal deaths:** Also decreased over the period, but at a slightly slower pace compared to other age categories, indicating persistent challenges in the earliest stages of life.
- **Comparison:** The three indicators followed nearly parallel downward trajectories, with under-five deaths consistently the highest and neonatal deaths the lowest, yet all converging toward significantly reduced levels by 2020.

(Figure 1. Comoros: Infant, Under-Five, and Neonatal Deaths, 1980–2020)

(Table 1. Pivoted dataset summary)

1.6 5. Interpretation

- The steady decline across all mortality indicators reflects major improvements in maternal and child healthcare, immunization coverage, and nutrition programs implemented since the 1980s.
- The sharper reduction in under-five and infant deaths highlights effective responses to preventable diseases and expanded access to community-based healthcare.
- The relatively slower decline in neonatal deaths suggests that birth-related complications and early-life infections remain significant challenges that require targeted interventions.
- Overall, these trends demonstrate Comoros’s progress toward Sustainable Development Goal 3.2, which aims to end preventable deaths of newborns and children under five.

1.7 6. Limitations

- National-level data may mask regional disparities, especially between urban centers and remote islands.
- Modeled WDI estimates are subject to uncertainty, particularly for earlier years with limited data coverage.
- The descriptive analysis does not identify specific policy impacts, funding sources, or health-care system reforms responsible for the observed decline.

1.8 7. Next Steps / Extensions

- Disaggregate mortality data by region, gender, and socioeconomic status to identify inequality in child health outcomes.
- Analyze links between health expenditure, immunization rates, and mortality reduction.
- Compare Comoros’s progress with that of other small island developing states (SIDS) to contextualize relative achievements and remaining challenges.

- Conduct a time-series regression to assess whether the decline in neonatal deaths has lagged significantly behind other categories, and what factors may explain the difference.

```
[1]: # How have infant, under-five, and neonatal deaths evolved relative to one
      ↪ another in Comoros between 1980 and 2020?

import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "health_com_filtered.csv" # Filtered dataset with only relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").
    ↪reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Comoros dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
plt.figure(figsize=(10,6))
plt.plot(df_plot["Year"], df_plot["Number of infant deaths"],
         marker='o', linestyle='-', label="Number of infant deaths")
plt.plot(df_plot["Year"], df_plot["Number of under-five deaths"],
         marker='o', linestyle='-', label="Number of under-five deaths")
plt.plot(df_plot["Year"], df_plot["Number of neonatal deaths"],
         marker='o', linestyle='-', label="Number of neonatal deaths")
```

```
plt.title("Comoros: Infant deaths vs Under-five deaths vs Neonatal deaths_↵
↵(number) (1980-2020)")
plt.xlabel("Year")
plt.ylabel("Number")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder,↵
↵"comoros_infant_deaths_vs_underfive_deaths_vs_neonatal_deaths.png"))
plt.show()

# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder,↵
↵"comoros_infant_deaths_vs_underfive_deaths_vs_neonatal_deaths"), index=False)
```

Pivoted Comoros dataset:

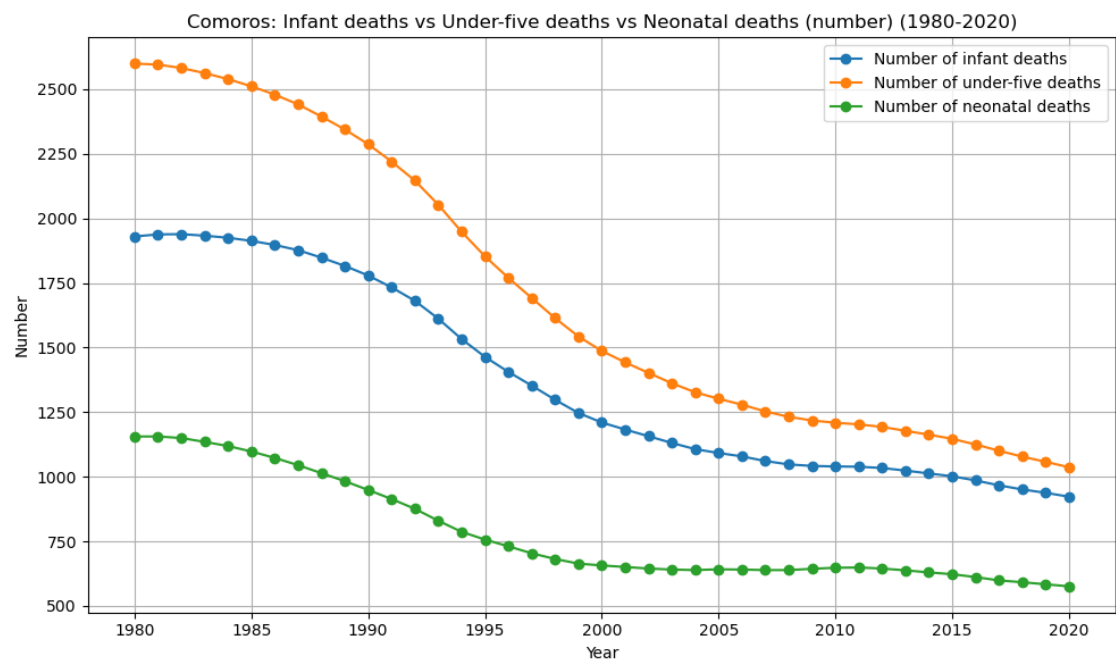
Indicator Name	Year	Number of infant deaths	Number of neonatal deaths	\
0	1980	1930	1156	
1	1981	1938	1156	
2	1982	1939	1150	
3	1983	1933	1135	
4	1984	1925	1119	
5	1985	1913	1098	
6	1986	1897	1073	
7	1987	1877	1045	
8	1988	1848	1014	
9	1989	1816	983	
10	1990	1779	949	
11	1991	1733	914	
12	1992	1681	876	
13	1993	1612	830	
14	1994	1533	787	
15	1995	1464	757	
16	1996	1406	731	
17	1997	1353	704	
18	1998	1298	682	
19	1999	1247	664	
20	2000	1210	657	
21	2001	1183	651	
22	2002	1157	645	
23	2003	1131	641	
24	2004	1107	639	
25	2005	1092	642	
26	2006	1079	641	
27	2007	1061	639	
28	2008	1048	639	
29	2009	1042	644	

30	2010	1040	648
31	2011	1039	649
32	2012	1034	645
33	2013	1024	638
34	2014	1013	630
35	2015	1002	623
36	2016	986	612
37	2017	967	599
38	2018	951	592
39	2019	938	584
40	2020	923	576

Indicator Name Number of under-five deaths

0	2599
1	2595
2	2582
3	2562
4	2539
5	2511
6	2479
7	2441
8	2394
9	2344
10	2287
11	2220
12	2146
13	2053
14	1948
15	1853
16	1770
17	1692
18	1614
19	1543
20	1487
21	1443
22	1402
23	1362
24	1327
25	1302
26	1279
27	1253
28	1232
29	1218
30	1209
31	1203
32	1193
33	1178
34	1163

35	1147
36	1125
37	1101
38	1078
39	1058
40	1037



[]: